DO NOW — RECALL

Search engine indexing and PageRank

Name: _____ Class: _____ Date: _____

1. What is a hyperlink?

2. What is an index in retrieval?

3. What kind of graph models links?

4. BST root20, left10, right30; insert 25. Where is it placed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A reference to another page/resource.

2. Model answer

A structure associating terms or keys with documents/records.

3. Model answer

A directed graph.

4. Model answer

25>20 go right to30; 25<30 go left of30.

Vocabulary — know these terms

crawler — Discovers/fetches documents.

inverted index — Maps terms to documents.

PageRank — Iterative link-based importance model.

damping — Probability factor in random-surfer model.

out-degree — Number of directed outgoing links.

NEW LEARNING

Read, annotate and explain

Understand

A crawler discovers pages by following links and sitemaps; an index records useful information about documents for searching.

Apply

PageRank models a random web surfer and passes importance through directed links. A link from a high-ranked page can carry more weight than one from a low-ranked page.

Evaluate

PageRank uses damping and repeated iterations; raw incoming link counts alone are not the full algorithm. Search engines combine many ranking signals.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

Predict first

Outgoing degree of A is two, so its rank is divided across two outgoing edges.

Explain

Both B and C receive half of A's contribution before damping and other link contributions.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Distinguish crawling and indexing. [4]

Question 2

In directed link graph A→B,C and B→C, identify incoming links of C and A's share given rank 0.4, ignoring damping. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why the number of inbound links alone is not an adequate PageRank calculation. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A page's PageRank is simply its number of incoming links.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.